

Week 4 Update — COVID-19 Global Data Analysis

Shrey Jain • Vaccination rollout: speed & coverage by country • Prepared for Neelam Gupta

Weekly update

This week I analyzed vaccination rollout across countries along two axes: **coverage** (share of population fully vaccinated) and **speed** (peak daily doses per 100 people, and days from a country's first dose to reaching 40% fully vaccinated). Analysis uses country-level rows and OWID income-group aggregates, in a reproducible Plotly notebook.

Discovered:

- **Coverage varies enormously** — from ~90%+ fully vaccinated (United Arab Emirates, Singapore, Hong Kong, Chile, Cuba, China) down to low single digits in parts of Sub-Saharan Africa (e.g. Burundi ~0.3%).
- **Speed is a separate axis from coverage**. Israel vaccinated fastest early (peak ~2.0 doses/100/day, 40% in ~78 days); some slower starters still reached high coverage later (e.g. Japan).
- **Income group is the dominant divide** — high-income countries pulled far ahead early, while low-income countries lagged by a wide margin throughout.

Focus set — speed & coverage metrics

Country	Fully vax %	1+ dose %	Peak doses /100/day	Days to 40%
Israel	65%	71%	1.96	78
United Arab Emirates	104%	106%	1.65	100
Chile	90%	92%	1.52	152
United Kingdom	75%	80%	1.44	145
United States	69%	81%	1.06	152
Germany	76%	78%	1.38	191
Japan	83%	84%	1.37	175
India	67%	72%	0.71	343
South Africa	35%	40%	0.40	n/a
Nigeria	37%	43%	0.23	n/a

Peak doses/100/day is the maximum 7-day-smoothed daily dose rate; "days to 40%" is n/a where a country never reached 40% fully vaccinated.

Charts

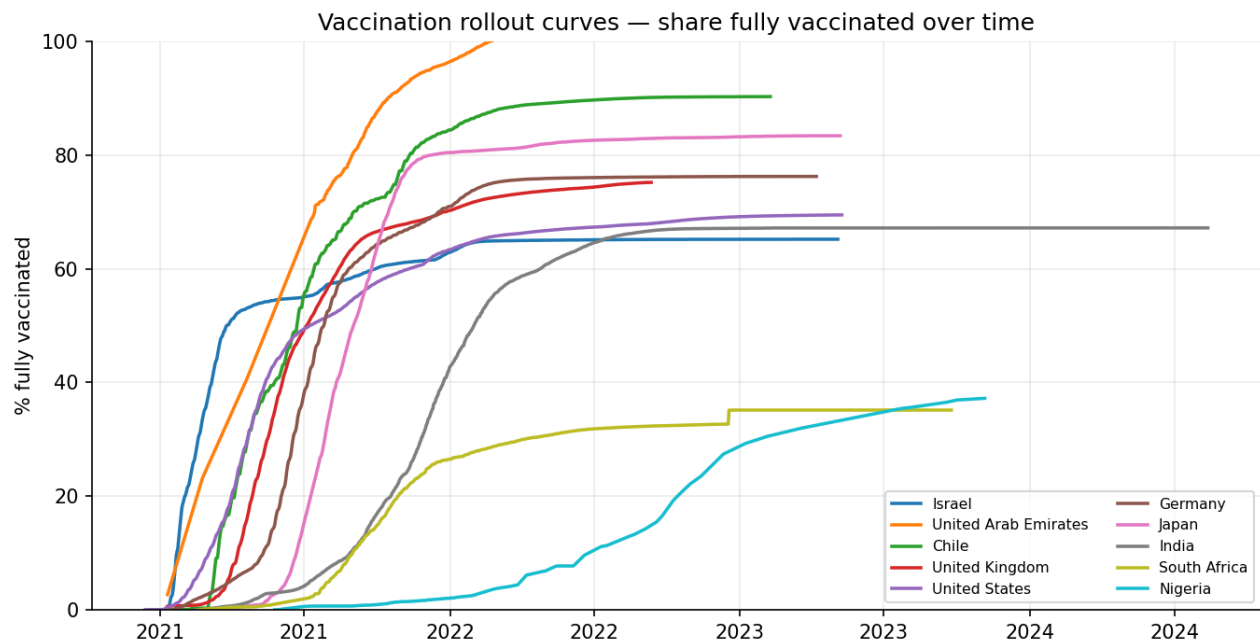


Fig 1. Rollout curves — steeper slope means faster rollout; the plateau height is final coverage.

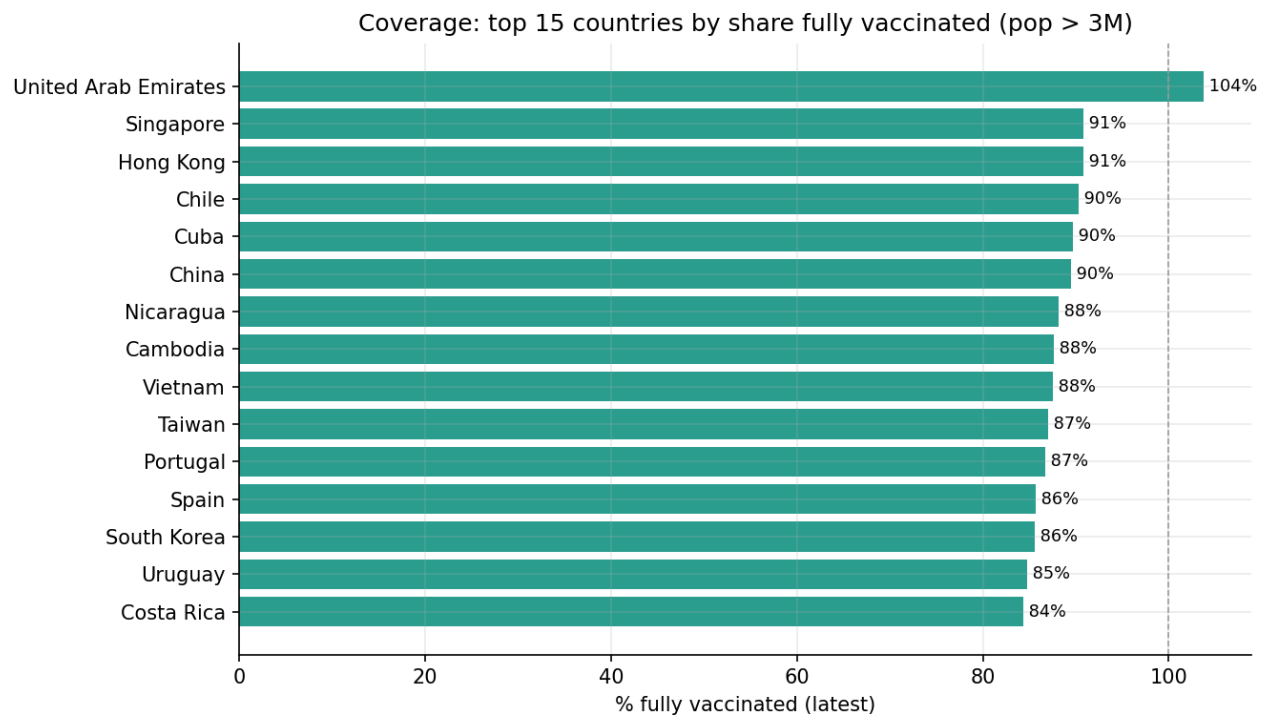


Fig 2. Coverage: the 15 highest countries by share fully vaccinated (population > 3M). Dashed line marks 100%.

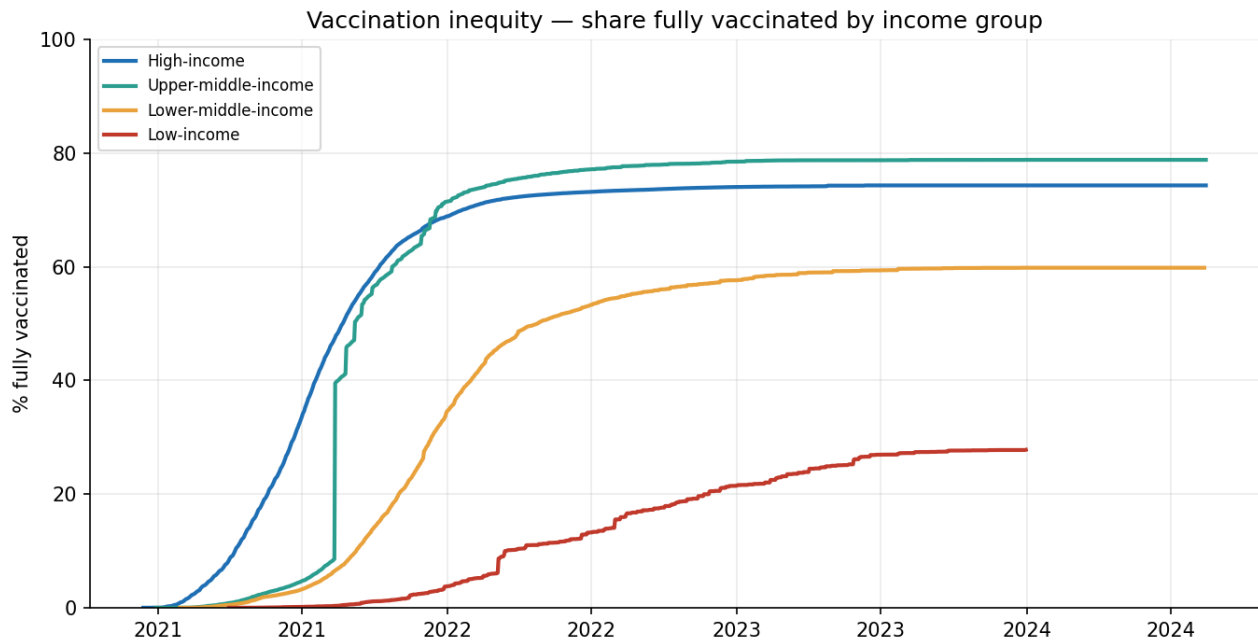


Fig 3. Coverage by income group — the widest and most persistent gap in the data.

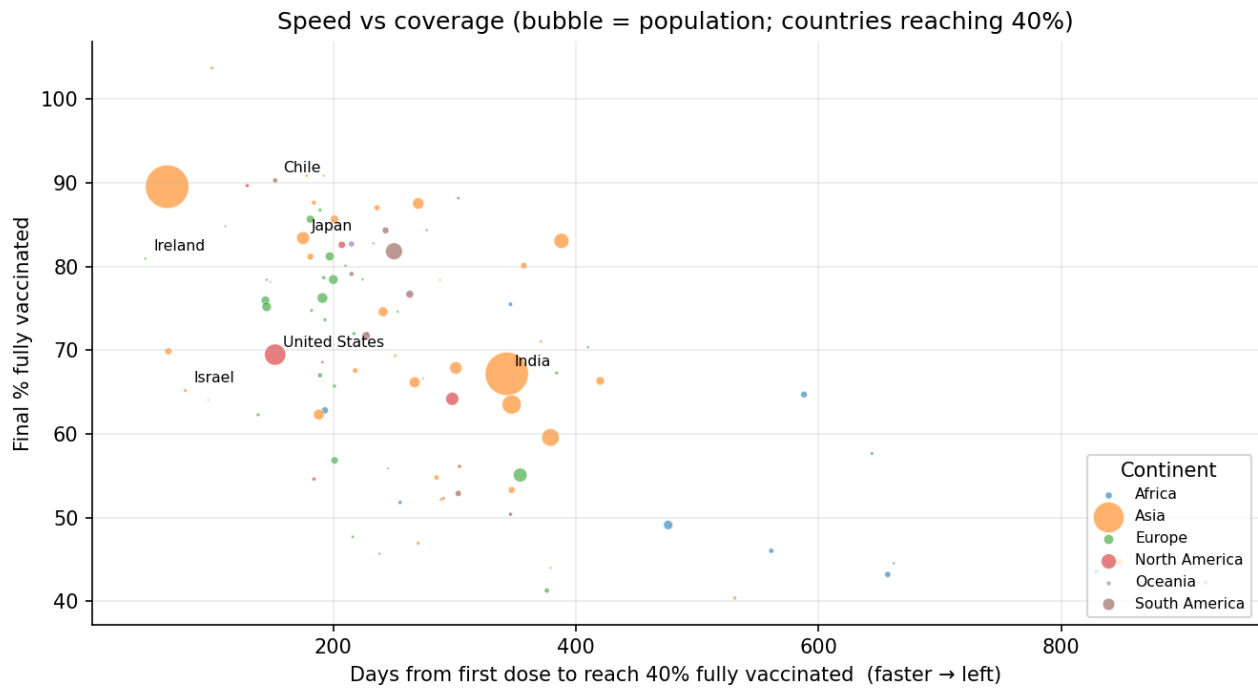


Fig 4. Speed (days to 40%, faster to the left) vs final coverage; bubble size is population.

Work artifacts & data sources

Repository	github.com/shreyjain99/covid19-global-analysis
Week 4 folder	github.com/shreyjain99/covid19-global-analysis/tree/main/week%204
Week 4 notebook	week4_vaccination_rollout.ipynb (interactive Plotly)

Charts	figures/fig1–fig4 (.png) + vax_table.csv
Data source	Our World in Data COVID-19 dataset (github.com/owid/covid-19-data)

Reproducibility (how to re-run)

- `pip install -r requirements.txt`.
 - Open `week4_vaccination_rollout.ipynb` and Run All — it downloads the dataset automatically and regenerates every figure from the raw data.
 - Or from the command line: `python scripts/analysis_week4.py (charts)` then `python scripts/build_week4_report.py` (this PDF).
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Risks / blockers & support needed

- **No blockers to delivery.** The week's objective is met and reproducible.
- **Data caveat:** per-hundred metrics can exceed 100% where doses are administered to non-resident / migrant populations (e.g. UAE), and rollout figures depend on national reporting cadence, so exact dates and rates carry some uncertainty.
- **Support:** confirm whether the 40% threshold is the preferred speed milestone, or another level (e.g. 70%) is expected for this comparison.